U-10 DILUTE INSULIN (10 UNITS/ML) CALCULATOR HELP

*The CHLA **DILUTE** Insulin Calculator will accurately determine the correct dose of **dilute (10 units/mL)** rapid-acting insulin to be administered based on a specific patient order, the measured blood glucose, and (IF carb-counting) the grams of carbohydrates eaten.*

*Dilute Insulin is generally used in very small patients who are very sensitive to insulin. Because Dilute insulin is only 1/10th the concentration of standard U-100 insulin, **doses are expressed in Syringe Units** based on markings on an insulin syringe.*

Insulin/Carbohydrate Ratio: (ABSENT IF CORRECTION-ONLY ORDER)

- *1 Syringe Unit of insulin covers **X** grams of carbohydrates eaten*
- Allowed range: 1.5 – 40 gm: 1 Syringe Unit

  ordered value; click to confirm

  indicates an ordered value that should be clarified

Correction Factor (CF) -- also called Insulin Sensitivity Factor (ISF):

- *1 Syringe Unit of insulin reduces a Blood Glucose elevated over Target Blood Glucose by **X** mg/dL*
- Allowed range: 10 – 200 mg/dL: 1 Syringe Unit

  ordered value; click to confirm

  indicates an ordered value should be clarified

Target Blood Glucose:

- *blood glucose target above which correction must be applied*
- Allowed range: 130 – 250 mg/dL
  - "Usual" Target BG (~10th – 90th %ile): 150 – 200

  ordered value; click to confirm

  indicates an extremely low or high value that should be clarified

    - LOW: < 130 | HIGH: > 250

****The calculator is not enabled for input of carbs eaten or measured blood glucose until all above parameters are confirmed by clicking the or***

Rapid-acting insulin in past 2 hrs?:

- *If yes, will not include any correction dose for high blood glucose in total dose calculation in order to prevent insulin stacking for insulin on board*
- Calculated dose will include carb coverage only
    - Defaults to YES if MAR shows rapid-acting insulin charted within past 2 hrs
    - Defaults to NO if recent dose not detected
        - No recent dose
        - Dose given but not charted (home, pump, other)
- response may be changed based on circumstance

Carbohydrates Eaten: (ABSENT IF CORRECTION-ONLY ORDER)

- *Enter the grams of carbohydrates eaten*
- calculator will accept any value from 0 – 99 gm

  ❌ unacceptable entry which must be corrected

Blood Glucose for Calculation:

- *Enter the current glucose measurement*
- calculator will accept any value from 1 – 2,000 mg/dL

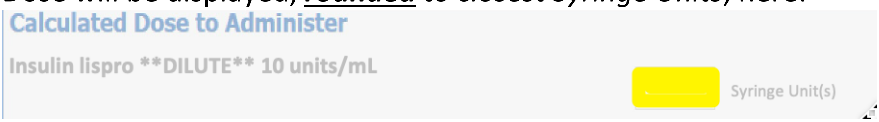
  ⚠ BG > 600 mg/dL; verify and correct if data entry was incorrect

  BG < 70 mg/dL: hypoglycemia
    - STOP – Calculator will not calculate dose
    - DO NOT GIVE INSULIN
    - requires rescue intervention for low BG

  ❌ unacceptable entry which must be corrected

To complete calculation of dose to administer:

- *Assure all parameters and patient values are valid*
- Click Calculate button
- Dose will be displayed, ***rounded*** *to closest Syringe Units*, here:



- Transfer calculated dose to MAR after independent 2nd witness check and dose has been administered
    - ***REMEMBER WHEN CHARTING***

- *Change time from default to time actually administered*
- *Change dose from **"1 dose"** to **"X Syringe Units"** actually administered*